

Human Motion Computing with Inertial Sensing

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1 Introduction

Human Motion Computing studies how human movement can be sensed, represented, analysed, and interpreted using computational methods. In this work, the KSAS dataset [1] is analysed using XROCKET [2], an explainable time-series representation method designed to extract discriminative temporal features from multivariate signals.

The objective of this report is to investigate not only whether the recorded karate movements can be classified, but also which sensor signals and temporal structures contribute most to the classification. The explanations provided by XROCKET are therefore analysed at three complementary levels: sensor channels, temporal dilations, and discriminative temporal patterns.

The analysis addresses the following questions:

Task 1.1: Which sensor signals contribute most to movement classification?

Task 1.2: Does the classifier rely mainly on short-duration patterns, long-duration global structures, or a combination of temporal scales?

Task 1.3: Which XROCKET patterns are most discriminative, and to what extent can they be interpreted from a human-motion perspective?

2 Dataset

The KSAS dataset [1] contains inertial recordings of American Kenpo Karate Blocking Set I. The dataset includes 20 participants performing five blocking movements with both arms, together with an additional absence-of-movement condition.

The six classes considered in this work are:

Table 1: Movement classes in the KSAS dataset.

Class	Movement
0	Absence of movement
1	Upward Block
2	Hammering Inward Block
3	Extended Outward Block
4	Outward Downward Block
5	Rear Elbow Block

Each class contains 40 recordings, giving a total of 240 examples. Each recording is represented by 18 channels obtained from six sensor types, with three axes per sensor:

- Accelerometer: x, y, z
- Gravity: x, y, z

- Gyroscope: x, y, z
- Linear acceleration: x, y, z
- Game rotation vector: x, y, z
- Magnetic field: x, y, z

The recordings have variable temporal length. In the dataset used for this analysis, sequence length ranges from 18 to 56 samples, with a mean length of 33.675 samples and a median length of 33 samples. No missing values were found during dataset inspection.

3 Methodology

3.1 Input Preparation

Each KSAS recording was loaded from its corresponding CSV file and represented as a multivariate time series with shape $(18, T)$, where 18 is the number of sensor channels and T is the number of time samples in the recording.

During loading, the recordings were checked to ensure that all sensor channels contained numeric values and that no missing values were present. Movement class, participant identity, and execution side were extracted from the filename of each recording.

The original temporal length of each recording was retained, resulting in variable-length multivariate time series provided directly to XROCKET.

3.2 XROCKET Configuration

XROCKET was used to transform each multivariate motion sequence into a fixed-dimensional feature representation. The encoder received the 18 sensor channels and used a maximum kernel span of 33 samples, corresponding approximately to the typical sequence length in the dataset.

First-order channel combinations were used, meaning that each XROCKET feature was associated with a single sensor channel. XROCKET used 84 fixed temporal kernels and four dilation values: 1, 2, 3, and 4.

The dilation determines the temporal spacing between the elements of a kernel and therefore the temporal scale covered by the resulting feature. With a kernel length of nine, increasing the dilation spreads the kernel elements over a larger portion of the time series. For dilations 1, 2, 3, and 4, the kernel covers 9, 17, 25, and 33 samples from its first to its last element, respectively. Lower dilation values therefore capture more local temporal patterns, whereas higher dilation values capture broader structures of the movement.

The main XROCKET parameters were:

- Input channels: 18
- Maximum kernel span: 33
- Combination order: 1
- Kernel length: 9
- Number of temporal kernels: 84
- Dilations: $\{1, 2, 3, 4\}$

For each cross-validation fold, a new XROCKET encoder was created. Its pooling thresholds were initialized from one recording belonging to the training set, after which the same encoder was used to transform all training and test recordings of that fold.

3.3 Participant-Independent Evaluation

To evaluate generalization to unseen individuals, a participant-independent five-fold cross-validation protocol was adopted using `GroupKFold`, with participant identity used as the grouping variable.

Since the dataset contains 20 participants, each fold used recordings from 16 participants for training and recordings from four different participants for testing. This ensured that recordings from the same participant could not appear in both the training and test sets of the same fold.

Within each fold, XROCKET was used to transform the training and test recordings into feature representations. A Random Forest classifier with 200 trees was then trained on the XROCKET features obtained from the training participants. The Random Forest was used as the main classifier for the explainability analysis because its feature importance values can be mapped back to the metadata of the XROCKET features, allowing sensor channels, temporal dilations, and discriminative patterns to be analysed directly.

Logistic Regression was also evaluated as a comparison model. For this classifier, the XROCKET features were standardized using statistics computed from the training fold only and the same transformation was then applied to the corresponding test fold.

Classification performance was evaluated using accuracy and macro-averaged F1 score. Final performance values were obtained by averaging the results across the five folds.

3.4 Explainability Analysis

The explainability analysis used the metadata associated with the XROCKET features together with the feature importances produced by the Random Forest classifier.

Sensor contribution was analysed by aggregating feature importances according to the corresponding input channel. Temporal-scale contribution was analysed by grouping feature importances according to dilation, allowing the relative importance of shorter and broader temporal structures to be compared.

The most discriminative features were identified using the joint combination of temporal pattern, dilation, and sensor channel. These combinations were ranked according to their mean Random Forest importance across the cross-validation folds.

To support their interpretation, the activations of the highest-ranked patterns were examined across movement classes. Test-set activations were standardized using statistics computed from the corresponding training fold, and the resulting values were compared to identify the movement classes associated with stronger or weaker pattern responses.

4 Results

4.1 Classification Performance

With the selected XROCKET configuration, each recording was transformed into a feature vector containing 6048 features.

Table 2 reports the participant-independent classification performance obtained across the five cross-validation folds.

Table 2: Participant-independent classification performance across five cross-validation folds. Values are reported as mean $\pm$ standard deviation.

Classifier	Accuracy	Macro-F1
Random Forest	0.871 ± 0.111	0.873 ± 0.107
Logistic Regression	0.888 ± 0.065	0.887 ± 0.063

Both classifiers achieved comparable mean performance across the five participant-independent folds. Logistic Regression obtained slightly higher average scores, reaching an accuracy of 0.888 and a macro-F1 score of 0.887, compared with 0.871 and 0.873 for Random Forest, respectively.

The relatively small difference between the two classifiers suggests that the XROCKET representation contains discriminative information that can be exploited by different classification models. At the same time, Random Forest showed greater variability across folds, with higher standard deviations for both accuracy and macro-F1. This indicates that its performance was less consistent across the different participant splits.

Overall, these results show that the XROCKET representation contains sufficient discriminative information to separate the six movement classes under a participant-independent evaluation protocol.

4.2 Task 1.1 – Sensor Channel Contribution

The mean Random Forest importance across the five participant-independent folds shows that the most influential individual sensor channels were:

Table 3: Most important sensor channels across the five participant-independent folds.

Channel	Mean importance	Std. dev.
game_rot_vec_x	0.070407	0.008654
game_rot_vec_y	0.069208	0.002638
accelerometer_x	0.065543	0.003501
game_rot_vec_z	0.063034	0.004251
magn_field_z	0.058987	0.011300

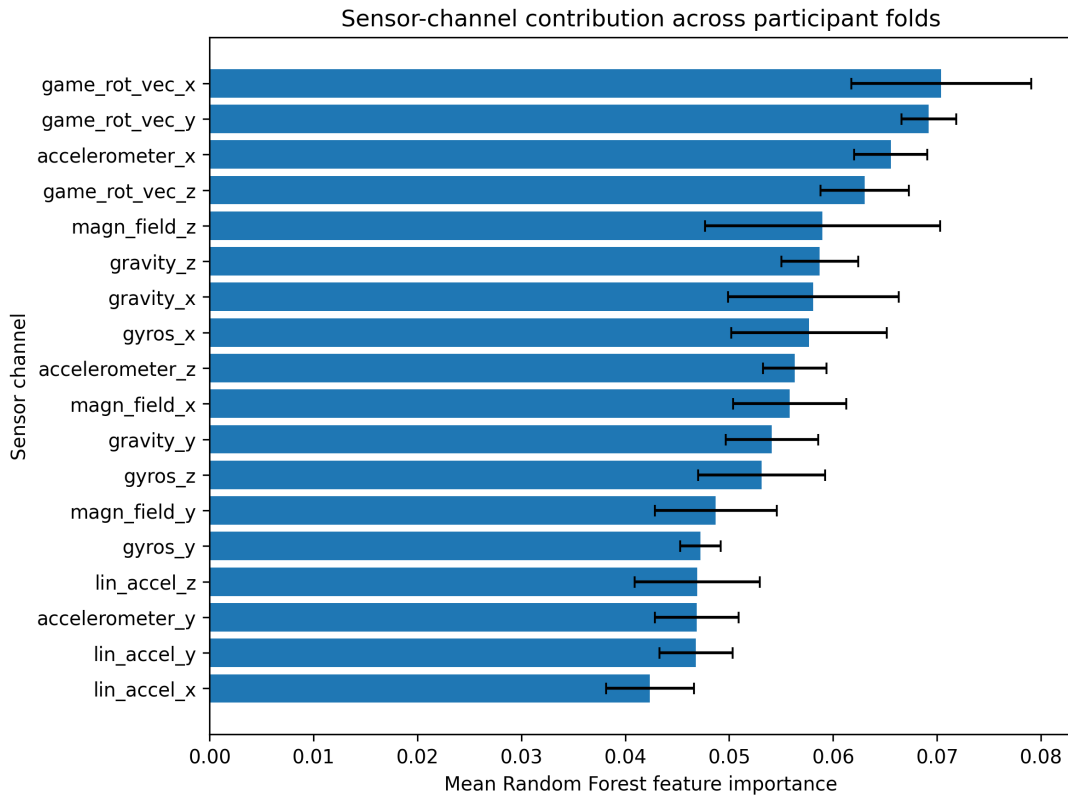


Figure 1: Mean Random Forest feature importance aggregated by sensor channel across the five participant-independent folds. Error bars represent the standard deviation across folds.

The channel importances are distributed across the different sensor signals, with no single channel clearly dominating the classification. However, the ranking shows a consistent contribution from the game rotation vector: its three components are all among the four highest-ranked channels. This suggests that orientation-related information is particularly useful for distinguishing the blocking movements. At the same time, the contribution of accelerometer, gravity, gyroscope, and magnetic-field channels shows that the classification also benefits from other characteristics of the recorded motion.

From a human-motion perspective, this is plausible because the five blocking techniques differ in the way the arm moves throughout the execution. Different trajectories, changes of direction, rotations, and acceleration profiles produce different inertial patterns over time. The results therefore suggest that XROCKET distinguishes the movements by combining information about orientation with information about the dynamics of the execution.

The x , y , and z components refer to the coordinate system of the sensing device [3]. Because the KSAS documentation does not specify the orientation of the smartphone relative to the participant’s arm, the individual axes cannot be mapped reliably to specific anatomical directions. The interpretation therefore focuses on the motion information captured by each sensor rather than assigning a specific anatomical meaning to these components.

4.3 Task 1.2 – Temporal Pattern Analysis

Mean Random Forest feature importance aggregated by dilation is reported in Table 4.

Table 4: Mean feature importance by dilation across the five participant-independent folds.

Dilation	Mean importance	Std. dev.
1	0.143461	0.025872
2	0.180126	0.005196
3	0.286290	0.022049
4	0.390122	0.008127

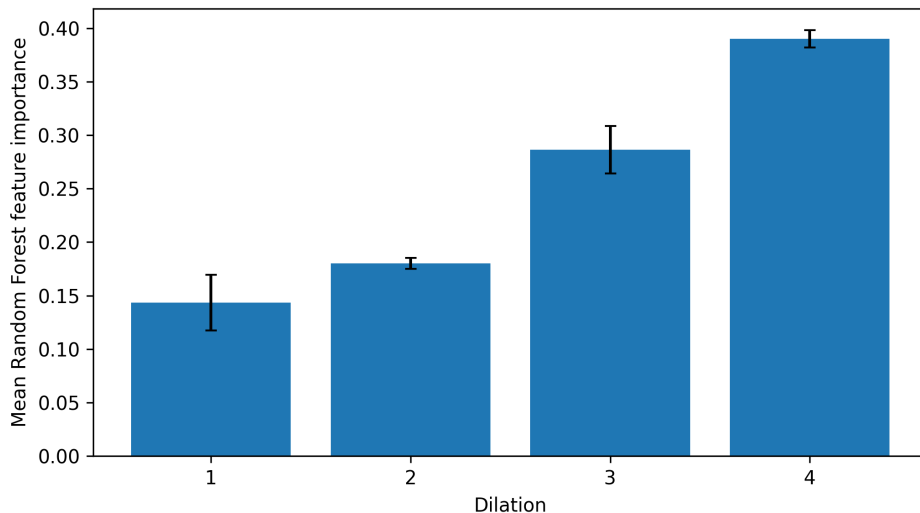


Figure 2: Mean Random Forest feature importance aggregated by dilation across the five participant-independent folds. Error bars represent the standard deviation across folds.

The importance increases with dilation, with dilation 4 showing the highest mean value. Dilations 3 and 4 account together for approximately 67.6% of the total importance, whereas dilations 1 and 2 account for the remaining 32.4%.

Higher dilation values spread the kernel elements over a larger portion of the time series and therefore capture broader temporal structures, while lower values focus on more local changes. The results indicate that the classification is multiscale, but places greater emphasis on the broader temporal scales.

This distribution can be related to the structure of the blocking movements. Broader temporal scales can capture how a technique evolves over a substantial part of its execution, whereas lower dilations can capture local transitions and faster changes. The distinction between movement classes therefore appears to depend mainly on their broader temporal evolution, while local motion patterns still provide complementary information.

4.4 Task 1.3 – Explainable Human Motion Computing

To identify the most discriminative XROCKET patterns, the importance assigned by the Random Forest to the generated features was analysed. Each feature can be associated with a temporal pattern, a sensor channel, and a dilation. The feature importances were therefore aggregated according to these three components and averaged across the five participant-independent folds.

Table 5 reports the five highest-ranked combinations.

Table 5: Five highest-ranked XROCKET pattern, channel, and dilation combinations.

Rank	Channel	Dilation	Mean importance	Std. dev.
1	game_rot_vec_y	4	0.006353	0.001559
2	game_rot_vec_z	4	0.006088	0.001416
3	gravity_z	4	0.005636	0.001213
4	game_rot_vec_x	4	0.005117	0.001492
5	game_rot_vec_x	3	0.005010	0.001416

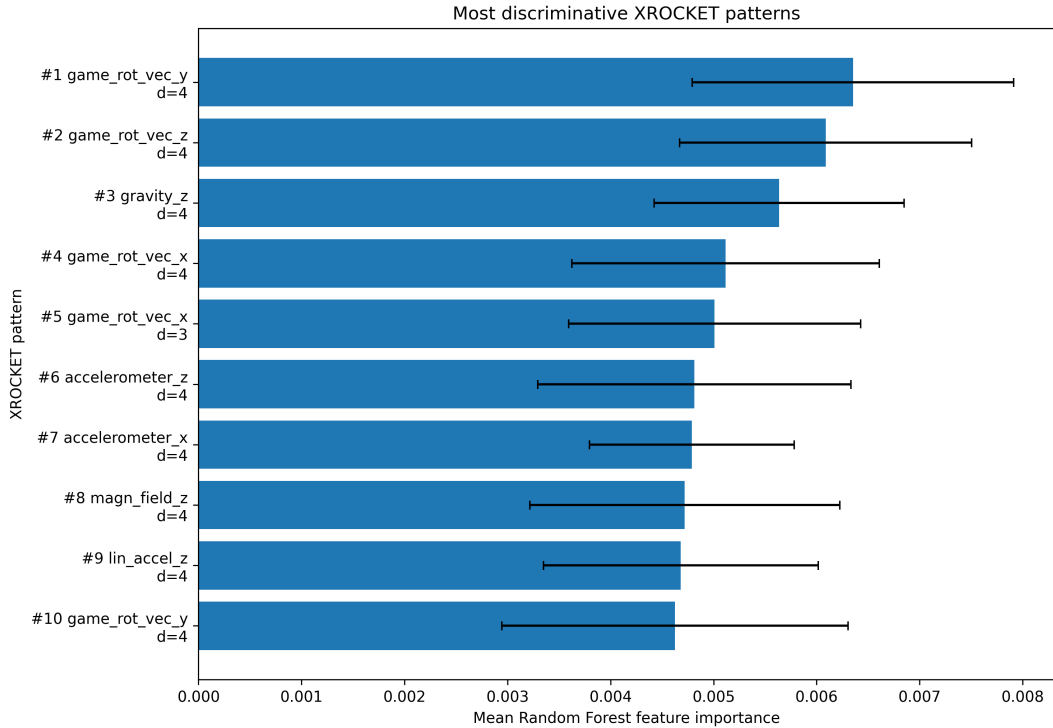


Figure 3: Highest-ranked XROCKET pattern, channel, and dilation combinations according to mean Random Forest importance across the five participant-independent folds.

The ranking is consistent with the previous analyses. Four of the five highest-ranked combinations involve game rotation vector channels, and four use dilation 4. This indicates that some of the strongest discriminative patterns are associated with changes in orientation-related signals occurring over relatively broad temporal scales.

Each pattern corresponds to a fixed XROCKET kernel applied to a specific sensor channel at a given dilation. Since the kernel coefficients do not have a direct biomechanical interpretation, the analysis focuses on the associated sensor signal, the temporal scale, and how the pattern response varies across movement classes.

To examine whether these patterns respond differently to the movements, their standardized activations were compared across classes. A positive value indicates a response above the corresponding training-fold average, while a negative value indicates a response below that average. Figure 4 shows the activation distribution of the highest-ranked pattern.

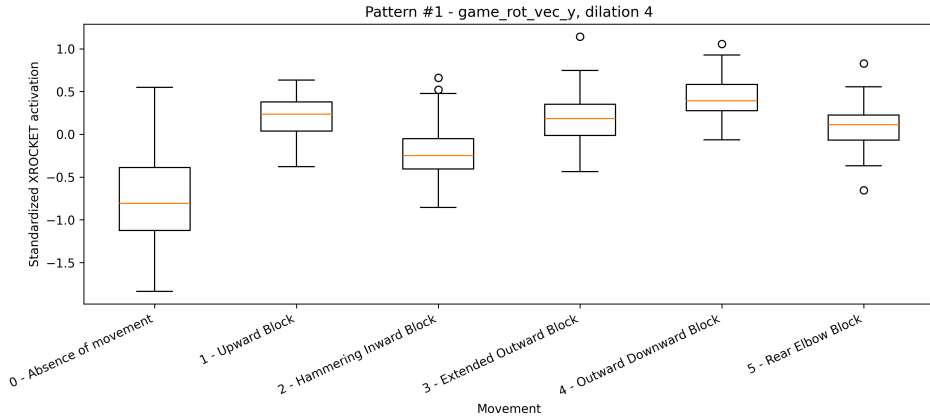


Figure 4: Standardized activation of the highest-ranked XROCKET pattern across the six movement classes.

The highest-ranked pattern operates on `game_rot_vec_y` with dilation 4. Its mean activation is highest for the Outward Downward Block and lowest for the absence-of-movement condition. The second-ranked pattern, based on `game_rot_vec_z` with the same dilation, shows the same highest and lowest classes. Patterns ranked third to fifth instead show their highest mean activation for the Extended Outward Block and their lowest for the absence-of-movement condition.

Several of the strongest patterns therefore show a clear contrast between active movement and inactivity. However, this is not the only distinction captured by the representation. Some of the twenty highest-ranked patterns also show different responses between active techniques. For example, one pattern based on `game_rot_vec_y` has its highest mean activation for the Extended Outward Block and its lowest for the Rear Elbow Block, while another has its highest activation for the Rear Elbow Block and its lowest for the Outward Downward Block. This suggests that XROCKET also captures temporal information that is useful for distinguishing the blocking techniques from one another.

Overall, the most important patterns suggest that the classification relies strongly on changes in orientation-related and other inertial signals over time. Their association with larger dilations indicates that broader temporal structures are particularly informative. These explanations provide a clearer view of the classifier’s behaviour by linking its decisions to the sensor signals involved, the temporal scale of the relevant changes, and the differences observed across movement techniques.

This information could also support movement learning. For example, the sensor responses of a learner could be compared with those of reference executions to identify differences in relevant signals or temporal patterns. Such differences could help provide more focused feedback on the execution.

However, KSAS contains movement-class labels and does not provide annotations of execution quality, correctness, or expertise. The present experiment therefore demonstrates explainable movement classification, but it does not directly demonstrate performance or skill assessment. Applying the same approach to that purpose would require data containing appropriate performance-related labels.

5 Discussion and Limitations

Overall, the results point in the same direction: orientation-related channels, especially the game rotation vector, are often among the most informative signals, while larger dilation values tend to receive greater importance. The most discriminative patterns combine these two aspects, suggesting that broader temporal changes in orientation-related and other inertial signals play an important role in distinguishing the blocking movements.

At the same time, lower dilations and other sensor families still contribute, so the representation remains multiscale and distributed across different types of motion information.

There are also some limitations to consider. The dataset contains 20 participants, and the Random Forest performance varies across the participant-independent folds. The reported results should therefore be interpreted in the context of this dataset and evaluation protocol.

The KSAS documentation does not specify the orientation of the smartphone relative to the participant’s anatomy. For this reason, the individual x , y , and z components cannot be mapped reliably to specific anatomical directions.

Random Forest feature importance indicates how useful a feature is for the classifier, but it should not be interpreted as a direct causal or biomechanical relationship. Similarly, the XROCKET patterns describe discriminative temporal structures in the sensor signals rather than explicit physical actions.

Finally, KSAS provides labels for the performed movement but not for execution quality, correctness, or expertise. The present analysis can therefore explain movement classification, but it cannot directly evaluate how well a technique was performed.

6 Generative AI Usage

Generative AI was used as a support tool during the development of this assignment. In particular, ChatGPT was used to assist with code debugging, organization of the analysis workflow and interpretation of intermediate outputs.

All experiments were executed locally by the author. Numerical results and figures were produced directly from the implemented analysis pipeline and all AI suggestions were reviewed before being included in the final work.

7 Conclusion

This work applied XROCKET to the KSAS inertial dataset to analyse movement classification from three complementary perspectives: sensor contribution, temporal scale, and discriminative temporal patterns.

The results showed that several sensor channels contribute to the classification, with a particularly strong contribution from the game rotation vector. The temporal analysis indicated that the representation is multiscale, although larger dilations received greater importance. The pattern analysis was consistent with these findings, since many of the highest-ranked combinations involved orientation-related channels and broader temporal scales.

Overall, XROCKET provided a useful representation for participant-independent movement classification while also allowing the classifier’s behaviour to be examined in terms of sensor signals and temporal structure.

8 Code Repository

The source code is available at:

https://github.com/francesco-pittacolo/human_motion_computing

References

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